

4NF examples

Subject	Teacher	Book
S1	T1	B1
S1	T2	B2
S2	T3	B3
S3	T4	B4

4NF examples

SID	SNAME	CID	CNAME
S1	A	C1	C
S1	A	C2	D
S2	B	C1	C
S2	B	C2	D

4NF examples

MOVIE	LOCATION	TYPE
M1	L1	T1
M1	L1	T2
M2	L2	T3
M2	L2	T4
M3	L3	T5

JOIN DEPENDENCY

- Join decomposition is a further generalization of Multivalued dependencies.
- If the join of R_1 and R_2 over C is equal to relation R then we can say that a join dependency (JD) exists, where R_1 and R_2 are the decomposition $R_1(A, B, C)$ and $R_2(C, D)$ of a given relations $R(A, B, C, D)$.
- Alternatively, R_1 and R_2 are a lossless decomposition of R .
- A JD $\bowtie \{R_1, R_2, \dots, R_n\}$ is said to hold over a relation R if $R_1, R_2, \dots, R_n$ is a lossless-join decomposition.
- The $*(A, B, C, D), (C, D)$ will be a JD of R if the join of join's attribute is equal to the relation R . Here, $*(R_1, R_2, R_3)$ is used to indicate that relation R_1, R_2, R_3 and so on are a JD of R .

5NF examples

- A single supplier can supply a product which will be used by more than one customers
- A customer can buy two or more products from a single supplier
- A product used by a customer can be supplied by more than one supplier

SUPPLIER	PRODUCT	CUSTOMER
S1	P1	C1
S1	P2	C2
S1	P1	C2
S2	P2	C1
S1	P2	C1

5NF examples

How to eliminate join dependency ?

- Join dependency is converted to separate tables
- The simplest solution to eliminate join dependency from the above table is to break the table into 3 binary one-one relationships

5NF examples

The resultant tables which are in 5th normal form are:

SUPPLIER	PRODUCT
S1	P1
S1	P2
S2	P2

SUPPLIER	CUSTOMER
S1	C1
S1	C2
S2	C1

PRODUCT	CUSTOMER
P1	C1
P2	C2
P1	C2
P2	C1

5NF examples

- When we combine these three tables, the resultant table will be the original table
- But this might not be true in all cases i.e., when we combine the separated tables, the resultant will not be equivalent to the original table, in that case we should not break down the original table.

5NF examples

Table – ACP

AGENT	COMPANY	PRODUCT
A1	PQR	Nut
A1	PQR	Bolt
A1	XYZ	Nut
A1	XYZ	Bolt
A2	PQR	Nut

5NF examples

Table – R1

AGENT	COMPANY
A1	PQR
A1	XYZ
A2	PQR

Table – R2

AGENT	PRODUCT
A1	Nut
A1	Bolt
A2	Nut

Table – R3

COMPANY	PRODUCT
PQR	Nut
PQR	Bolt
XYZ	Nut
XYZ	Bolt

5NF examples

Natural join:

$R1 \bowtie R3 = R13$ (over COMPANY)

$R13 \bowtie R2 = R$ (over AGENT AND PRODUCT)

5NF examples

Agent	Company	Product_Name
Suneet	ABC	Nut
Raj	ABC	Bolt
Raj	ABC	Nut
Suneet	CDE	Bolt
Suneet	ABC	Bolt

agent_company_product_table

The assumption being that ABC makes both Nuts and Bolts and that CDE makes Bolts only. This table can be decomposed into its three projections without loss of information as demonstrated below:

P1		P2		P3	
Agent	Company	Agent	Product_Name	Company	Product_Name
Suneet	ABC	Suneet	Nut	ABC	Nut
Suneet	CDE	Suneet	Bolt	ABC	Bolt
Raj	ABC	Raj	Bolt	CDE	Bolt
		Raj	Nut		

5NF examples

All redundancy has been removed, if the natural join of P1 and P2 is taken, the result is:

Agent	Company	Product_Name
Suneet	ABC	Nut
Suneet	ABC	Bolt
Suneet	CDE	Nut*
Suneet	CDE	Bolt
Raj	ABC	Bolt
Raj	ABC	Nut

The spurious row is asterisked. Now, if this result is joined with P3 over the column 'company' the following table is obtained:

Agent	Company	Product_name
Suneet	ABC	NUT
Suneet	ABC	BOLT
Suneet	CDE	BOLT
Raj	ABC	BOLT
Raj	ABC	NUT

5NF examples

SUPPLIER	PRODUCT	CUSTOMER
S1	P1	C1
S1	P2	C2
S2	P3	C1
S2	P4	C3
S3	P1	C2
S3	P4	C3
S3	P2	C3